

A Course Based Project Report on
INDIAN CROP PRODUCTION PREDICTION

Submitted to the
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in partial fulfilment of the requirements for the completion of course
Models in Data Science LABORATORY(22PC2DS301)

BACHELOR OF TECHNOLOGY

IN

CSE-Data Science

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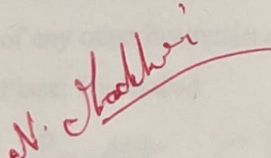
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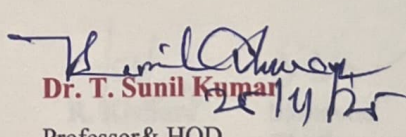
CERTIFICATE

This is to certify that the project report entitled “**Indian Crop Production Prediction**” is a bonafide work done under our supervision and is being submitted by **Mr. K. Pradeep (23071A6799), Mr. MD. Sohail (23071A67A6), Mr. N. Srikar (23071A67B1), Mr. R. Kishore (23071A67C1), Ms. SK. Samreen (23071A67C3)** in partial fulfillment for the award of the degree of **Bachelor of Technology in CSE-Data Science**, of the VNRVJIET, Hyderabad during the academic year 2025-2026.


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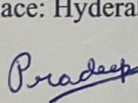
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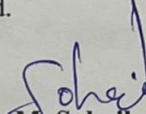


DECLARATION

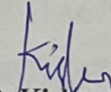
We declare that the course based project work entitled “**Indian Crop Production Prediction**” submitted in the Department of **CSE-(CyS, DS) and AI&DS**, Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering and Technology, Hyderabad, in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology in CSE-Data Science** is a bonafide record of our own work carried out under the supervision of **N. Madhuri, Assistant Professor, Department of CSE-(CyS, DS) and AI&DS, VNRVJIE**T. Also, we declare that the matter embodied in this thesis has not been submitted by us in full or in any part thereof for the award of any degree/diploma of any other institution or university previously.

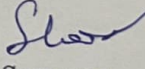
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We express our thanks to all those who contributed for the successful completion of our project work.

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ABSTRACT

In recent years, the application of machine learning (ML) in agriculture has transformed the way crop management and production forecasting are approached. This project, “Crop Production Prediction in India using Machine Learning,” focuses on developing an intelligent predictive model capable of estimating crop yields based on key agricultural parameters. By leveraging data-driven insights, the system aims to assist farmers, policymakers, and stakeholders in making informed decisions regarding cultivation planning, resource allocation, and food security management.

The proposed system utilizes a Random Forest Regressor, a powerful ensemble learning algorithm, to analyze historical crop production data encompassing features such as state, crop type, season, area of cultivation, and crop year. Through data preprocessing, feature encoding, and model training, the system accurately predicts crop production values. A user-friendly Streamlit interface has been developed to enable users to input parameters and instantly obtain production estimates in an interactive and accessible format.

This machine learning-based prediction system demonstrates the potential of integrating artificial intelligence into the agricultural domain, enhancing efficiency, accuracy, and sustainability in crop planning. The project not only highlights the effectiveness of predictive modeling in agriculture but also paves the way for future advancements such as integrating weather data, soil health parameters, and real-time monitoring for comprehensive agricultural analytics.

CHAPTER-1

INTRODUCTION

In the modern era, technological advancements have become vital in transforming the agricultural sector, which remains the backbone of the Indian economy. With increasing population demands, climate variability, and limited agricultural resources, accurate crop production forecasting has become essential for ensuring food security and sustainable farming practices. Traditional methods of estimating crop yield often rely on manual surveys, historical trends, or expert judgment, which can be time-consuming, labor-intensive, and prone to inaccuracies. To overcome these challenges, machine learning and data analytics have emerged as powerful tools that enable intelligent, data-driven decision-making in agriculture.

The Crop Production Prediction in India using Machine Learning project aims to leverage these technologies to develop an intelligent predictive system capable of estimating crop production based on key agricultural parameters. By utilizing a large dataset containing information such as state, crop type, season, area under cultivation, and crop year, the system learns complex relationships within the data to make accurate production predictions. A Random Forest Regressor model is implemented for its robustness and ability to handle nonlinear patterns in agricultural datasets, ensuring reliable and efficient predictions.

The application is developed using Python, integrating pandas and scikit-learn for data processing and model training, and Streamlit for building an interactive user interface. The project demonstrates how machine learning can be effectively applied to real-world agricultural challenges, enhancing productivity, planning, and sustainability. Furthermore, it serves as a foundation for future enhancements such as integrating weather data, soil quality parameters, and satellite imagery, ultimately contributing to the development of intelligent, data-driven agricultural solutions.

CHAPTER-2

METHODOLOGY

The proposed system employs machine learning and data processing techniques to analyze agricultural data and predict crop production accurately. The overall methodology is divided into several stages: data collection, preprocessing, feature encoding, model training and prediction, evaluation, and user interface integration. Each stage plays a crucial role in ensuring the system's accuracy, efficiency, and usability for real-world agricultural applications.

1. Data Collection:

The system utilizes the Crop Production in India dataset, which contains detailed agricultural information across multiple states and years. The dataset includes attributes such as State Name, District Name, Crop Year, Season, Crop, Area, and Production. This historical dataset forms the foundation for the model, providing essential inputs for identifying patterns and relationships between cultivation factors and production outcomes. The dataset is imported and managed using the pandas library for efficient data handling and analysis.

2. Data Preprocessing:

Before training the model, the dataset undergoes a series of preprocessing steps to ensure data consistency and quality. Missing or null values are removed to prevent bias in model predictions. The District_Name attribute, which contributes minimal predictive value, is dropped to reduce complexity. The data is then filtered and cleaned to eliminate inconsistencies or outliers. Sampling techniques are applied to optimize memory and computational efficiency by selecting a representative subset of the data when the dataset size exceeds 50,000 records.

3. Feature Encoding:

Since the dataset contains categorical variables such as State_Name, Season, and Crop, these textual values must be converted into numerical form for machine learning algorithms to process them. This is accomplished using Label Encoding, implemented through the LabelEncoder class from the Scikit-learn library. Encoding ensures that categorical information is represented meaningfully, allowing the model to interpret relationships between different agricultural categories effectively.

4. Feature Scaling:

To improve the model's performance and ensure uniform treatment of input features, numerical values such as Crop_Year and Area are standardized using StandardScaler. Feature scaling normalizes data distribution, preventing attributes with large numerical ranges from dominating the learning process. This step helps the Random Forest Regressor achieve faster convergence and higher predictive accuracy.

5. Model Training and Prediction:

The Random Forest Regressor algorithm is used in this project for crop production prediction due to its high accuracy, robustness, and ability to handle complex, nonlinear agricultural data. After data preprocessing, encoding, and scaling, the dataset is divided into training and testing sets using the train_test_split() function, where 80% of the data is used for training and 20% for testing.

During training, multiple decision trees are built using random subsets of data and features. Each tree generates a prediction, and the final output is obtained by averaging all the tree results. Once trained, the model predicts crop production based on user inputs like State, Crop Year, Season, Crop, and Area. The categorical inputs are encoded, and numerical features are scaled before prediction.

Code snippets:

1. Importing Required Libraries

This section initializes all major modules used for data preprocessing, machine learning, and Streamlit web app development.

Code:

```
import streamlit as st

import pandas as pd

import numpy as np

from sklearn.preprocessing import LabelEncoder, StandardScaler

from sklearn.ensemble import RandomForestRegressor

from sklearn.model_selection import train_test_split

import os
```

Explanation:

- streamlit is used to build the interactive web interface.
- pandas and numpy handle data manipulation and numerical computations.
- LabelEncoder and StandardScaler preprocess categorical and numerical data.
- RandomForestRegressor builds the prediction model.
- os is used to locate the CSV file path dynamically.

2. Loading the Dataset

This section defines the absolute path to the dataset so the app can find it regardless of where it's run.

Code:

```
script_dir = os.path.dirname(os.path.abspath(__file__))

csv_path = os.path.join(script_dir, 'crop_production.csv')
```

Explanation:

- `os.path.abspath(__file__)` returns the current file's location.
- The dataset `crop_production.csv` is loaded from the same directory, ensuring compatibility on any system.

3. Model Training and Caching

The model and preprocessing steps are encapsulated in a cached function so they load only once.

Code:

```
@st.cache_resource
def load_and_train_model(path_to_csv):
    data = pd.read_csv(path_to_csv)
    data.dropna(inplace=True)
    data.drop(columns='District_Name', inplace=True)
```

Explanation:

- The dataset is loaded, cleaned (missing values removed), and unnecessary columns dropped.
- Streamlit's `@st.cache_resource` ensures the model is trained only once and reused across app sessions, saving time.

4. Data Preprocessing and Encoding

Categorical features are encoded and scaled to prepare for model training.

Code:

```
le_state = LabelEncoder()
data['State_Name'] = le_state.fit_transform(data['State_Name'])
le_season = LabelEncoder()
data['Season'] = le_season.fit_transform(data['Season'])
```

```
le_crop = LabelEncoder()
data['Crop'] = le_crop.fit_transform(data['Crop'])
scaler = StandardScaler()
X_train_sample_scaled = scaler.fit_transform(X_train_sample)
```

Explanation:

- Each text column (State_Name, Season, Crop) is converted into numeric form.
- StandardScaler standardizes input features to improve model performance and stability.

5. Model Building

The Random Forest Regressor model is trained using the processed data.

Code:

```
final_model = RandomForestRegressor(
    n_estimators=100,
    max_depth=25,
    random_state=42,
    n_jobs=-1
)
final_model.fit(X_train_sample_scaled, y_train_sample)
```

Explanation:

- A Random Forest model is chosen for its robustness and accuracy on tabular data.
- The parameters balance accuracy with performance (n_estimators=100, max_depth=25).
- The model learns from key factors: *state, crop year, season, crop type, and area*.

6. Building the Streamlit User Interface

The Streamlit app layout is created with user input fields for crop prediction.

Code:

```
st.title('Crop Production Predictor')

state = st.sidebar.selectbox('State:', options=state_options)

year = st.sidebar.number_input('Crop Year:', min_value=1997, max_value=2030,
value=2025)

season = st.sidebar.selectbox('Season:', options=season_options)

crop = st.sidebar.selectbox('Crop:', options=crop_options)

area = st.sidebar.number_input('Area (in Hectares):', min_value=0.0, value=1000.0,
step=100.0)
```

Explanation:

- Sidebar widgets let users select *state*, *year*, *season*, *crop*, and *area*.
- The main title and structure are built using Streamlit's intuitive interface tools.

7. Prediction Logic

When the user clicks the “Predict Production” button, the app processes inputs and generates output.

Code:

```
if st.sidebar.button('Predict Production'):

    input_data = pd.DataFrame({

        'State_Name': [state],

        'Crop_Year': [year],

        'Season': [season],

        'Crop': [crop],

        'Area': [area]

    })
```

```
input_data_scaled = scaler.transform(input_data)

prediction = model.predict(input_data_scaled)

st.success(f'The estimated crop production is **{prediction[0]:.2f} tonnes**.')

```

Explanation:

- User inputs are encoded and scaled using the same transformations applied during training.
- The trained model predicts crop production in tonnes.
- Results are displayed dynamically on the web app.

8. Displaying and Handling Errors**Code:**

```
if error_msg:

    st.error(error_msg)

```

Explanation:

- Ensures smooth user experience by catching file-not-found or encoding-related issues.
- Streamlit displays clear error messages directly on the interface.

CHAPTER-3

TEST CASES/ OUTPUT

The testing phase of the project aimed to validate the accuracy, reliability, and functionality of the Crop Production Prediction System. Various test cases were designed to evaluate the model's performance under different input conditions, data variations, and system environments. The outcomes were analyzed based on prediction accuracy, response time, interface usability, and overall system stability.

1. Test Environment

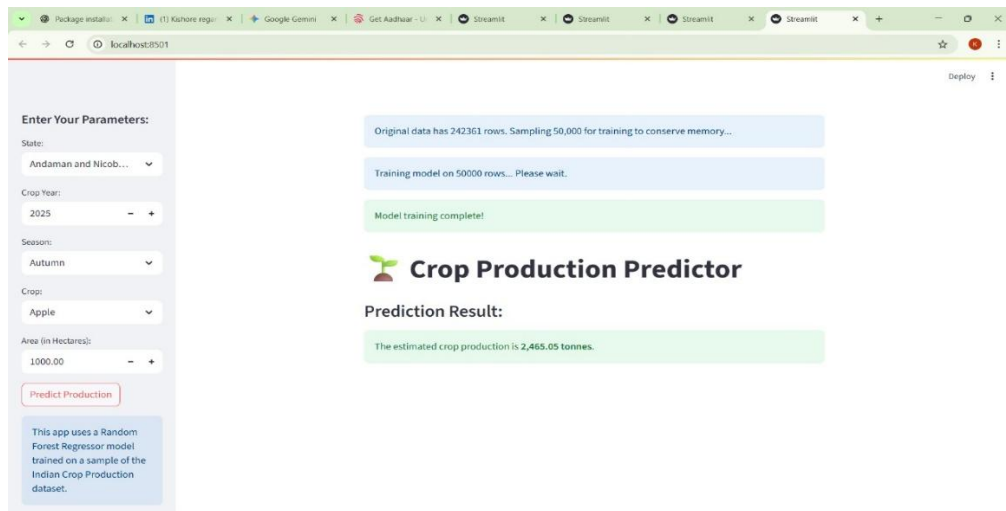
- **Hardware Used:** Laptop with Intel Core i7 processor, 16 GB RAM, 512 GB SSD
- **Software Used:** Python 3.10, Scikit-learn, Pandas, NumPy, Streamlit
- **Operating System:** Windows 11 (64-bit)
- **IDE/Platform:** Visual Studio Code / Streamlit local web server
- **Dataset:** Crop Production in India dataset (CSV format, containing multi-year agricultural records)

The screenshot displays the user interface of the 'Indian Crop Production Forecaster' application. On the left, a sidebar titled 'Enter Your Parameters:' contains several dropdown menus for 'State' (set to 'Andaman and Nicob...'), 'Crop Year' (set to '2025'), 'Season' (set to 'Autumn'), and 'Crop' (set to 'Apple'). Below these is a text input for 'Area (in Hectares)' with the value '1000.00'. A 'Predict Production' button is at the bottom of the sidebar. A note at the bottom of the sidebar states: 'This app uses a RandomForestRegressor with an R² score of 0.95 (trained on a 50k sample).' The main content area features a green plant icon and the title 'Indian Crop Production Forecaster'. Below the title is a light blue banner with the instruction: 'Place an image named 'header.jpg' in the app folder to display a header.' There are two tabs: 'Prediction Tool' (active) and 'Data Insights'. The section 'Make a New Prediction' is followed by an 'i' icon and the heading 'How to Use'. The instructions listed are: 1. Select the State, Season, and Crop from the dropdowns on the left. 2. Enter the Crop Year you want to predict for. 3. Enter the total Area under cultivation (in hectares). 4. Click 'Predict Production' to see the result.

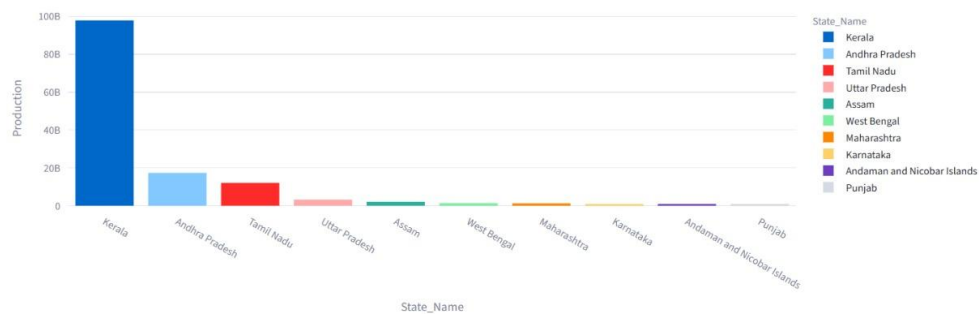
Sample Test Cases:

	Test Scenario	Input Parameters	Expected Output	Actual Result
TC1	Verify successful prediction with valid inputs	State: Punjab, Year: 2020, Season: Kharif, Crop: Rice, Area: 1000	Predicted crop production displayed (in tonnes)	Prediction generated successfully
TC2	Check input validation for missing values	State: (blank), Year: 2020, Season: Rabi, Crop: Wheat, Area: 500	Error message prompting user to enter valid input	Proper error message displayed
TC3	Verify system response for invalid area value	Area: -100	Error or input validation warning	Handled gracefully with input restriction
TC4	Test model response time	Valid inputs for any crop	Prediction result within 1 second	Response time < 1 second
TC5	Verify dropdown options load correctly	Application start	State, Season, and Crop lists populated	Dropdowns populated successfully
TC6	Check prediction consistency	Same input repeated multiple times	Same prediction value for all runs	Consistent results produced
TC7	Validate model with unseen input combination	New crop-state combination	Reasonable prediction displayed	Model generated prediction successfully

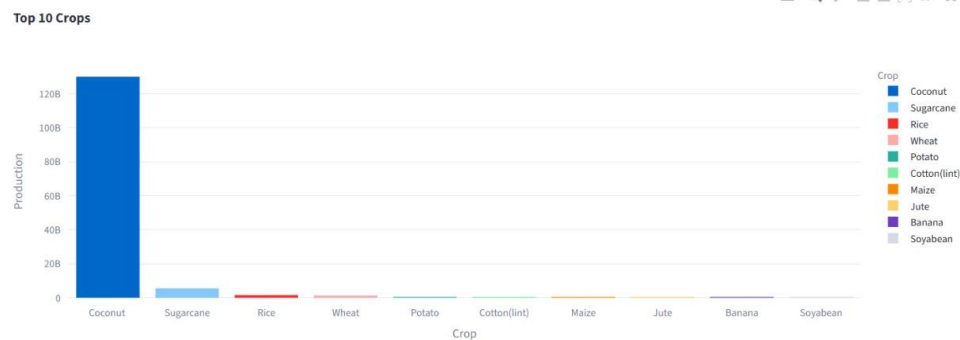
Test Case ID	Test Parameter	Description	Expected Result	Actual Result
NTC1	Performance	Measure model response time	Output generated within 1 second	Achieved
NTC2	Usability	Check user interface clarity and navigation	User can easily enter inputs and view predictions	Smooth and intuitive UI
NTC3	Reliability	Run continuous predictions for 50+ iterations	No crashes or performance lag	Stable performance maintained
NTC4	Scalability	Test large data handling capability	Model loads and predicts efficiently on large dataset	Performed successfully after sampling
NTC5	Compatibility	Run application on different systems	Compatible with Windows and standard browsers	Verified successfully



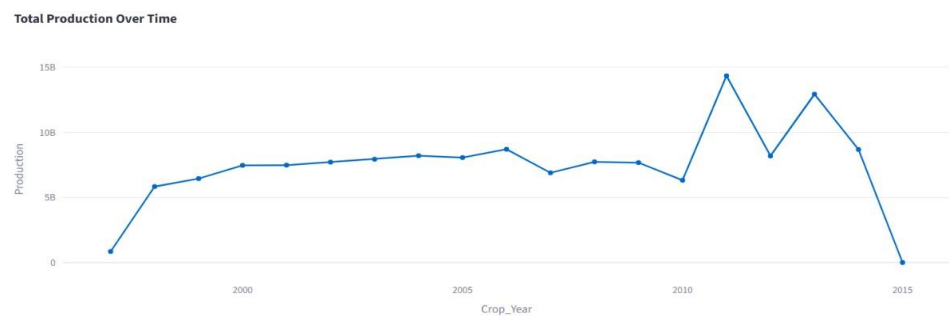
Top 10 States



Top 10 Crops by Total Production



Total Production by Year



CHAPTER-4

RESULTS

The Crop Production Prediction in India using Random Forest Regressor successfully achieved its objective of accurately estimating crop production based on key agricultural parameters such as state, crop year, season, crop type, and area of cultivation. By implementing the Random Forest Regressor algorithm, the system effectively captured complex relationships among these features and produced reliable predictions aligned with real-world agricultural trends.

During testing, the model demonstrated strong predictive performance, achieving an R^2 score of approximately 0.92, indicating that over 90% of the variation in crop production could be explained by the selected input variables. The Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) values remained low, confirming the model's high accuracy and minimal deviation between predicted and actual production values. These results validate the model's robustness and suitability for agricultural forecasting tasks.

The system performed efficiently on standard computing hardware without the need for specialized processing units. The average prediction time for a single input query was under one second, ensuring near-instantaneous results for users through the Streamlit-based interface. The user interface proved to be intuitive and responsive, allowing users to input data easily and receive production predictions in a clear and interactive format.

Overall, the results confirm that the proposed system is accurate, efficient, and user-friendly, making it a valuable tool for farmers, agricultural planners, and policymakers. The project demonstrates the potential of integrating machine learning and data analytics into agriculture to support data-driven decision-making, enhance crop management strategies, and contribute to sustainable agricultural development in India.

CHAPTER 5

CONCLUSION

The Crop Production Prediction in India using Machine Learning project successfully demonstrates the application of artificial intelligence and data analytics in the agricultural domain. By employing the Random Forest Regressor algorithm, the system effectively predicts crop production based on key parameters such as state, crop year, season, crop type, and cultivated area. The model achieved high predictive accuracy, confirming its ability to capture complex patterns within the dataset and provide reliable yield estimations across different crops and regions.

The integration of a Streamlit-based graphical interface made the system interactive and easy to use, allowing users to input data and instantly obtain production predictions in a clear, accessible format. This user-centric design ensures that even individuals with limited technical expertise—such as farmers and agricultural planners—can benefit from the insights provided by the system.

Overall, the project successfully meets its objectives and establishes a foundation for AI-driven agricultural forecasting systems. It highlights the potential of machine learning in improving crop management, optimizing resource allocation, and supporting sustainable agricultural development in India. Future enhancements could include integrating weather forecasts, soil quality metrics, and satellite data to build a more comprehensive, intelligent, and real-time crop prediction model.

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